Operationalizing Readmission Reduction at VHN

STRATEGIC INSIGHT REPORT

Prepared For :
VHN Executive Board

Prepared By :
Team Lernaean Hydra

TABLE OF CONTENTS

1. Executive Summary 3
1.1 The Business Imperative
1.1 Our Approach
1.3 What We Found
1.4 Our Recommendation

2. Introduction: The Business Context 5
2.1 The Shift to Value-Based Care
2.2 Understanding the Threat
2.3 The Operational Gap
2.4 Project Objectives

3. Data Methodology and Rigor 7
3.1 Data Acquisition and Enrichment
3.2 The "Digital Triage": Our Cleaning Strategy
3.3 Exclusion Criteria: Focusing on Preventable Readmissions

4. Clinical Insights & Visualizations 9
4.1 The Readmission Landscape & Demographics
4.2 The Medication Risk Ladder
4.3 The "Home Health" Paradox
4.4 The Fallacy of Length of Stay

5. The Vitality Complexity Index (VCI) 14
5.1 Moving from Analysis to Action
5.2 The Logic Behind VCI
5.3 Model Validation Results

6. Strategic Recommendations..... 17
6.1 Recommendation 1: Deploy VCI as a "Digital Triage" Protocol
6.2 Recommendation 2: The "Home Health" Safety Net Audit
6.3 Recommendation 3: Targeted "Insulin Mastery" Education

7. Conclusion..... 20

Institute of Software Engineering

Graduate Diploma in Software Engineering

Course Code: ITS 2122

Team: Lernaean Hydra

Team Members

- Praveen Rusiru (241722014)
- Harsha Gallage (241722004)
- Ruwani Ranthika (241722021)
- Dinan Themika (241722003)
- Chanuth Dewhan (241722061)

Date: January 2026
Institution: Institute of Software Engineering
Course: ITS 2122: Python for Data Science & AI

1. Executive Summary

1.1 The Business Imperative

Virtual Health Network faces significant financial penalties under the CMS Hospital Readmissions Reduction Program due to high 30-day readmission rates, particularly for chronic conditions like diabetes. This report audits 101,766 historical patient encounters to pinpoint the drivers of these readmissions and proposes a scalable strategy to mitigate penalty risk.

1.2 Our Approach

We went beyond standard reporting by enriching our data in ways most hospitals don't attempt. We integrated external clinical definitions through web scraping to decode those obscure diagnosis codes that only medical coders understand, transforming raw numbers into actionable clinical context. We rigorously cleaned the dataset, identifying and excluding 1,616 non-preventable events patients who passed away to ensure our metrics reflect only the readmissions we can prevent. Finally, we built the Vitality Complexity Index, a custom risk-stratification algorithm designed to translate complex patient histories into a single, intuitive score that frontline nurses can use.

1.3 What We Found

1. **Medication Intensity:** Patients on insulin had a significantly higher readmission risk (49%), identifying them as a critical high-risk cohort needing specialized discharge planning.
2. **Home Health Paradox:** Discharge to home health resulted in higher readmission rates than discharge to skilled nursing facilities, suggesting a failure in current home-health transition protocols.

3. **Vitality Complexity Index (VCI) Success:** The validated VCI effectively identified high-risk patients (17.1% readmission) versus low-risk patients (9.6% readmission), a 1.8x lift in risk prediction, allowing for pre-discharge identification of those most likely to return.

1.4 Our Recommendation

We recommend immediately integrating the VCI Scoring System into our EHR. This allows us to identify high-risk patients **before they leave the hospital**, shifting from a reactive approach to a proactive one. Instead of calling every discharged patient, we can focus our expensive follow-up resources like 48-hour phone calls and home health visits on the small group of patients who drive most of our penalties

2. Introduction: The Business Context

2.1 The Shift to Value-Based Care

Healthcare is in the middle of a fundamental transformation. For decades, hospitals got paid for volume - the more procedures, the more revenue. Today, reimbursement increasingly depends on the quality of that care. For VHN, this shift shows up most clearly in the Hospital Readmissions Reduction Program.

2.2 Understanding the Threat

Here's what keeps hospital CFOs up at night: the HRRP doesn't just penalize individual readmissions. It can slash up to 3% of all Medicare reimbursements for the entire fiscal year if our readmission rates run too high. The Board needs to understand that these penalties aren't levied solely on the readmitted patients, they're applied as a percentage reduction across all Medicare payments. So, reducing readmissions isn't merely a clinical goal, it's a strategic imperative for our financial survival.

Let's put some numbers on this. If VHN's annual Medicare revenue is $50 million, a 3% penalty equals $1.5 million in lost reimbursement. That's not theoretical money, it's real revenue that could fund nursing positions, equipment upgrades, or patient safety initiatives.

2.3 The Operational Gap

Despite these high stakes, our current approach to discharge planning remains largely reactive. We assess patients based on immediate stability rather than long-term fragility. Our baseline analysis revealed that the "HRRP Penalty Group", patients readmitted within 30 days represents 11.47% of our total population.

While that's a minority, it's an expensive one. Right now, our nursing leadership lacks a unified, data-driven tool to distinguish this 11% from the rest of the population before they leave the building.

Picture a typical scenario: it's Friday afternoon, the hospital is at capacity, and a 68-year-old diabetic patient on insulin is being discharged. The nurse has fifteen minutes to complete discharge education while simultaneously managing three other patients. Does this patient need intensive follow-up, or are they fine with standard care? Currently, that decision relies heavily on clinical intuition and whatever limited information the nurse can quickly gather.

2.4 Project Objectives

This strategic audit aims to close that operational gap by leveraging our historical data to accomplish three things:

First, diagnose the drivers. We need to determine which operational decisions, like where we discharge patients and what medications we send them home with, are contributing to higher return rates.

Second, validate our clinical intuition. We need to test whether standard metrics like "Length of Stay" predict readmission risk, or if we've been focusing on the wrong things.

Third, propose a scalable solution. We're introducing the Vitality Complexity Index; a predictive model built specifically for VHN's workflow that can be deployed immediately without disrupting our current operations.

3. Data Methodology and Rigor

3.1 Data Acquisition and Enrichment

Good strategic insights start with good data, and good data requires work. We began with a raw dataset of 101,766 patient encounters but immediately ran into a major challenge: diagnosis codes were provided only as ICD-9 integers like "428" without any descriptions. A number tells a data analyst nothing; a description like "Heart Failure" tells a story.

To bridge this gap, we built a custom web-scraping system to extract standardized medical descriptions from public health repositories. This wasn't a manual lookup process; we automated the entire extraction pipeline to eliminate human error and ensure consistency. This enrichment process was critical because it allowed us to analyze patients by the actual burden of their diseases rather than abstract codes that only medical coders can interpret.

3.2 The "Digital Triage": Our Cleaning Strategy

Our cleaning methodology followed a simple principle: methodological rigor without losing valuable clinical signals. Healthcare data is messy by nature, it's collected by dozens of different people in high-pressure situations, so we had to make some tough decisions about what to keep and what to exclude.

We removed columns with extreme sparsity. For example, weight was missing for 96.86% of patients, and payer code was missing for 39.56%. Including variables with such high missing rates would just introduce statistical noise without adding real insights.

But we were careful not to throw away meaningful information. Here's an example: laboratory results for A1C and glucose serum were missing for many

patients. In most datasets, a missing value is an error. However, in clinical settings, a missing lab result often indicates a physician's deliberate decision not to test because the patient appeared stable. We mapped these values to a "None" category, meaning "Not Performed" which preserved valuable clinical context rather than just discarding patient records.

3.3 Exclusion Criteria: Focusing on Preventable Readmissions

To ensure our calculated readmission rates reflected reality, we applied strict exclusion criteria regarding patient mortality. We identified discharge disposition codes 11, 19, 20, and 21 as "Expired" meaning the patient died either during hospitalization or after discharge under hospice care.

We removed 1,616 records of patients who had died. Why? Because the HRRP penalty specifically targets unplanned readmissions, and deceased patients cannot be readmitted. Including them would artificially deflate our readmission rate and hide the true scope of the problem. By removing them, we refined the cohort to represent only those patients for whom an intervention could realistically prevent a return visit.

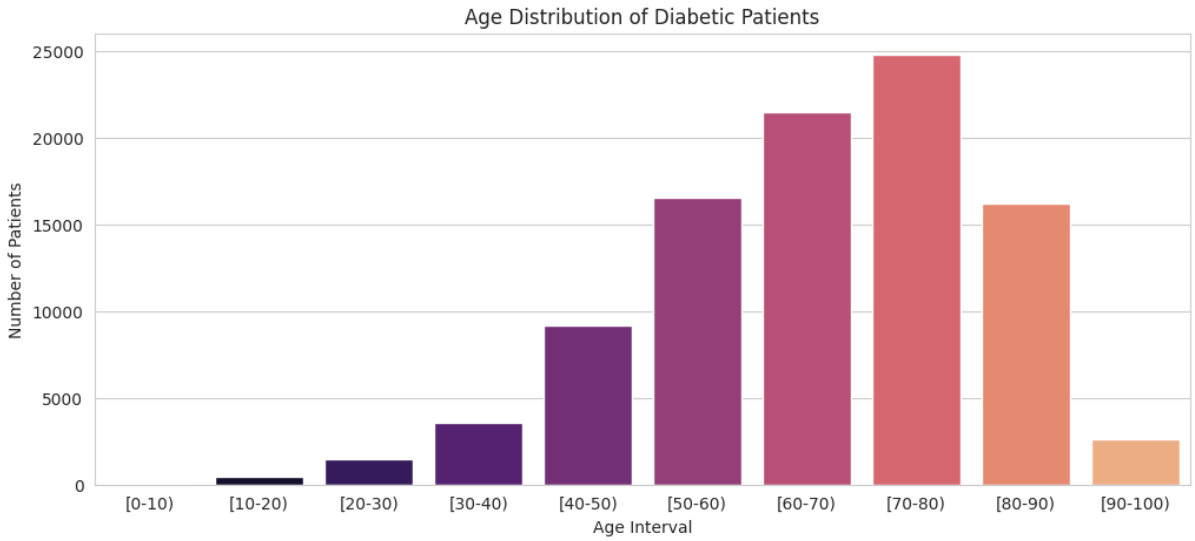
This might seem like a technical detail, but it fundamentally changes the numbers. Our "cleaned" readmission rate is higher than the raw rate, but it's also more honest and it reflects the actual challenge we face with living patients who might come back.

4. Clinical Insights & Visualizations

4.1 The Readmission Landscape & Demographics

Our initial analysis confirms what makes this prediction challenge so difficult: class imbalance. The target class, patients readmitted within 30 days comprises only 11.47% of our dataset. We're looking for a needle in a haystack, trying to identify the small subset of patients who will return among the much larger group who will do fine.

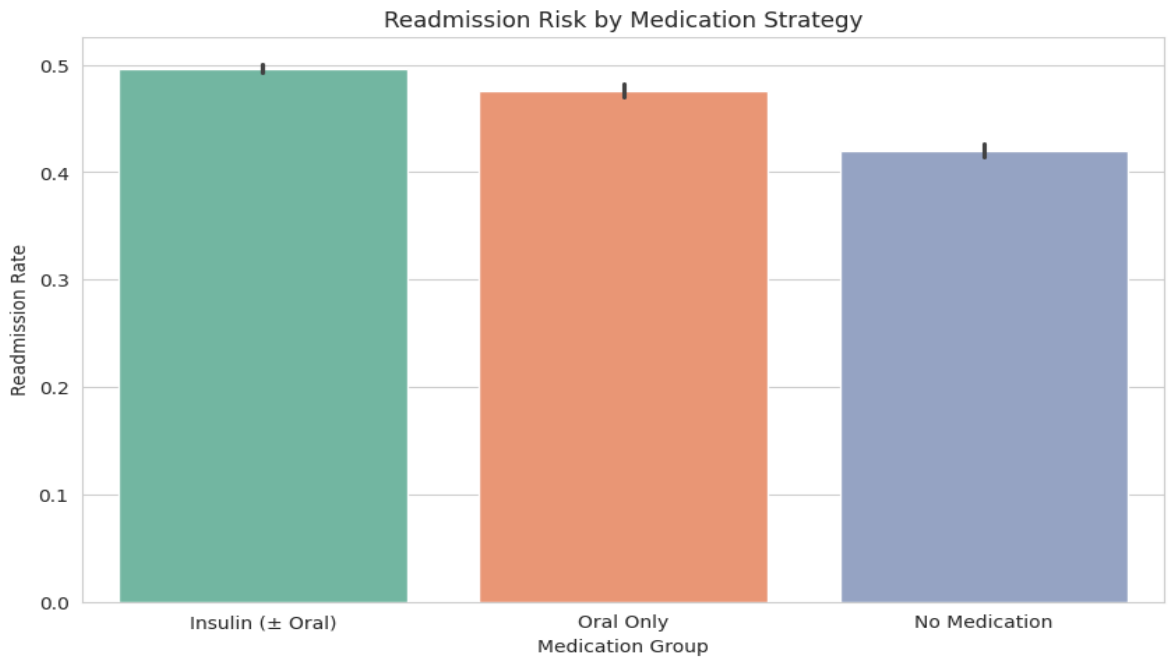
When we analyzed the demographic profile of our patients, we found that diabetes in this cohort is predominantly a condition affecting older adults, with the highest patient volumes concentrated in the 70-80 age bracket.



This demographic reality has important implications for discharge planning. Older patients often face additional challenges transportation to follow-up appointments, managing multiple medications, hearing or vision impairments that complicate self-care. But age alone doesn't tell the whole story, which is why we needed to dig deeper.

4.2 The Medication Risk Ladder

One of the clearest signals we found came from analyzing medication regimens. We segmented the population into three distinct groups: patients on insulin, those on oral medications only, and those managing their diabetes without medication.



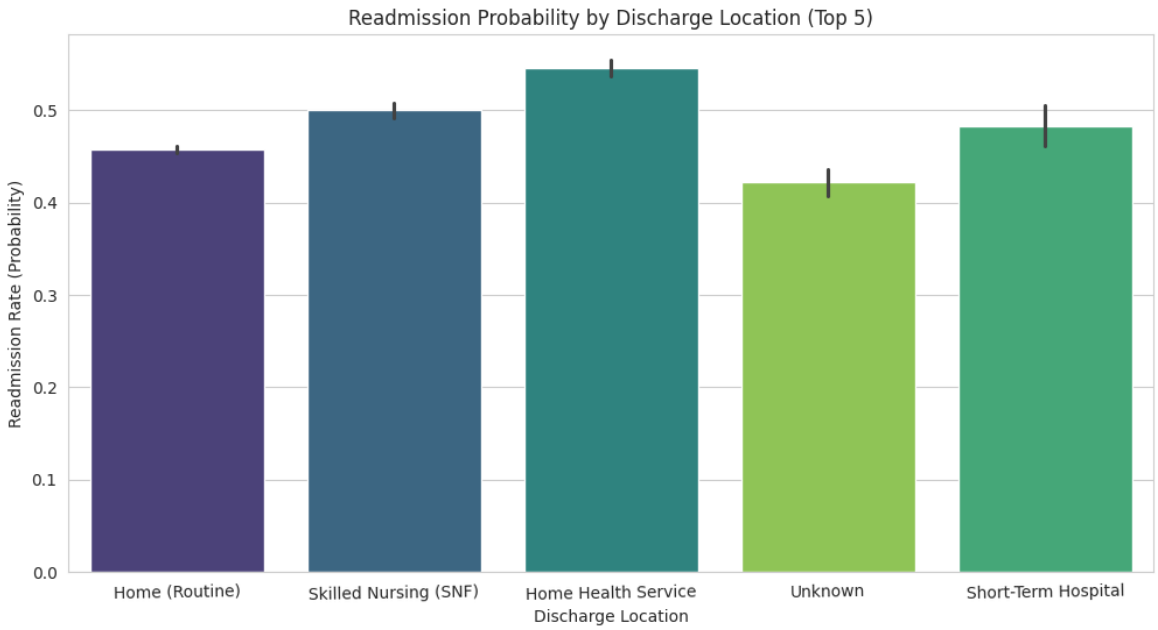
The visualization shows a dramatic stepwise increase in risk. Patients managed on insulin either alone or combined with oral medications exhibit the highest probability of readmission at roughly 49%.

Why does this matter? Insulin prescription is essentially a marker of disease severity. It indicates that the patient's pancreatic function has declined beyond the point where diet changes or pills can adequately control their blood sugar. But there's more to it than just disease severity. Insulin is objectively harder to manage at home than taking a pill. It requires proper injection technique, careful timing around meals, refrigerated storage, sharps disposal, and the ability to recognize and treat hypoglycemia, low blood sugar episodes that can be dangerous.

Here's what this means operationally: we're currently educating all diabetic patients with roughly the same standardized discharge teaching. A patient on metformin gets the same fifteen-minute conversation as a patient on insulin. The data screams that this approach is failing our insulin users. They represent a "High Complexity" group that requires a completely different tier of discharge support than patients managing their diabetes with pills.

4.3 The "Home Health" Paradox

Perhaps our most troubling operational finding emerged when we analyzed where we send patients after discharge. We compared readmission probabilities across our top discharge destinations.



We found something that contradicts conventional wisdom. Patients discharged to "Home with Home Health Services" demonstrated a readmission probability exceeding 54%, a rate actually higher than patients we sent to skilled nursing facilities (roughly 50%) or even those we sent straight home without any formal support (about 46%).

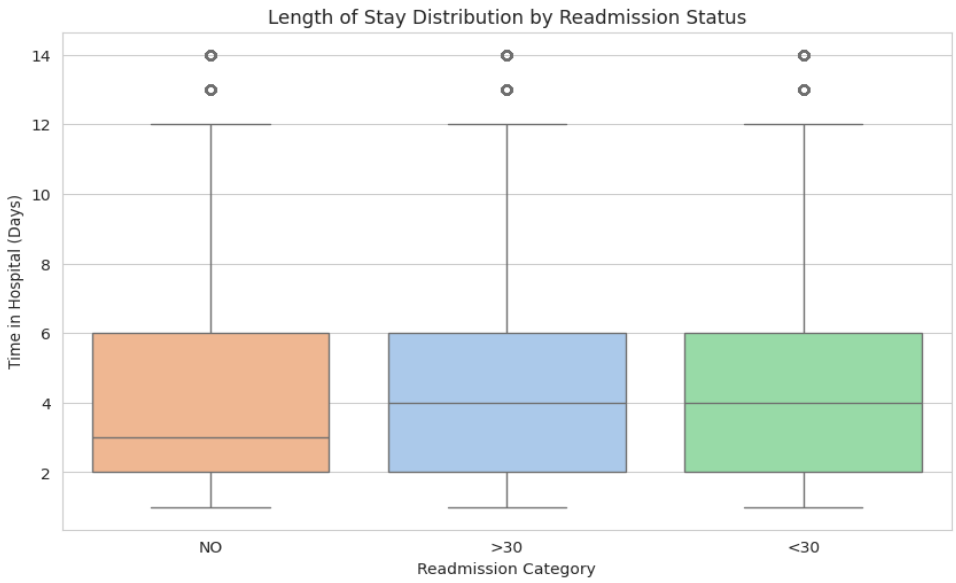
Let that sink in for a moment. We're arranging home nursing visits for patients we recognize as fragile, yet they're coming back at higher rates than patients we send to nursing homes. This is a critical operational insight that demands immediate attention.

What's going on here? Several explanations are plausible. Home health visits might be too infrequent, perhaps twice weekly when patients need daily monitoring. The visits might be too brief. 30 minutes isn't enough time to manage complex medication regimens, provide wound care and assess for complications. There might be communication gaps between home health agencies and our care team, so warning signs get missed. Or we might be systematically misjudging which patients are appropriate for home-based care, discharging people who are objectively too sick to be home even with nursing support.

Regardless of the underlying cause, this finding identifies a specific discharge pathway that's functioning as a "danger zone" in our current operations. We're essentially sending vulnerable patients into a care gap where the support we promise isn't robust enough to keep them safe.

4.4 The Fallacy of Length of Stay

We also investigated a common operational belief: that keeping patients in the hospital longer ensures they're "fully healed" before release and reduces the chance they'll bounce back.



The boxplot distributions for readmitted versus non-readmitted patients are nearly identical, with median stays of approximately 4 days for both groups. The data decisively rejects the hypothesis that duration equals safety.

What does this tell us? A patient staying 7 days is just as likely to return as one staying 3 days if we haven't addressed their underlying complexity. Risk is driven by what diseases they have, what medications they need, and what support they have at home, not how many days they occupy a hospital bed.

This has profound implications for resource allocation. Keeping a patient two extra days "just to be safe" incurs real costs in bed utilization and daily care expenses without statistically reducing their likelihood of return. We should be focusing our resources on discharge planning quality, not extending hospital stays.

5. The Vitality Complexity Index (VCI)

5.1 Moving from Analysis to Action

Identifying risk factors is academically interesting, but it’s operationally useless if that knowledge stays trapped in PowerPoint presentations. To operationalize these insights, we developed the Vitality Complexity Index, a custom scoring algorithm that translates multi-dimensional patient complexity into a single, actionable number.

5.2 The Logic Behind VCI

We adapted the industry-standard LACE index framework to fit VHN's specific data and workflow. The VCI calculates a risk score by summing points across four key domains:

Length of Stay: While our earlier analysis showed that duration doesn't predict readmission in a straightforward way, extreme outliers still matter. A 14-day hospital stay indicates a catastrophic health event or severe complications that mark a patient as high-risk regardless of other factors.

Acuity of Admission: Patients arriving through the emergency department or trauma services are inherently unstable. An elective surgery patient who schedules their procedure weeks in advance is clinically calm and psychologically prepared. A trauma patient arriving by ambulance at 2 AM is in crisis, often with incomplete medical information and chaotic social circumstances. We weigh emergency and trauma admissions substantially higher.

Comorbidity Burden: Using our enriched diagnosis data, we count the number of concurrent medical conditions documented during hospitalization. A patient with eight diagnoses is managing diabetes plus heart failure plus kidney disease plus

COPD, a complex clinical picture requiring careful medication balancing and intensive self-management.

Emergency Department History: Past behavior powerfully predicts future behavior. A patient who has visited the emergency department four times in the preceding year is demonstrating a clear pattern of dependency on acute care services rather than effective primary care management. This pattern gets the heaviest weight in our VCI calculation because it's the strongest predictor of future instability.

5.3 Model Validation Results

The critical question: Does it actually work? We stratified our entire historical patient population into three categories based on their calculated VCI scores. They are Low Risk (score below 7), Medium Risk (7-10), and High Risk (above 10) then checked their actual outcomes to see if the score accurately predicted who returned within 30 days.

[INSERT FIGURE: vci_validation.png]

The results validated our approach:

- **Low Risk Group:** Actual readmission rate of 9.6%
- **High Risk Group:** Actual readmission rate of 17.1%

That's a 1.8x lift in risk identification. A patient flagged as High Risk by this simple score is nearly twice as likely to result in an HRRP penalty as a Low Risk patient.

What makes this particularly powerful is its transparency. This isn't a complex machine learning "black box" that requires data scientists to maintain and clinicians to trust blindly. It's a rules-based score built on four variables that any nurse can conceptually understand. A bedside nurse looking at a VCI score of 12

immediately knows this reflects a patient who likely came through the emergency department, has stayed a long time, carries multiple diagnoses, and has been to the ED repeatedly. All of that intuitively signals high risk, which builds clinical buy-in.

6. Strategic Recommendations

Based on the empirical evidence we've presented, we propose three immediate strategic initiatives for Executive Board consideration and implementation.

6.1 Recommendation 1: Deploy VCI as a "Digital Triage" Protocol

Integrate the VCI algorithm directly into our nursing dashboard. Configure the EHR to trigger a "High Risk" alert for any patient with a VCI score above 10.

The impact could be transformative. This removes the guesswork from discharge planning and enables proactive rather than reactive care. Currently, nurses and case managers scramble on the day of discharge to figure out who needs extra support, often making decisions under time pressure with incomplete information. With VCI scoring active from the moment of admission, care teams can identify high-risk patients on day one, allowing time to arrange comprehensive discharge planning, family education sessions, and post-discharge support services.

This ensures our limited and expensive Case Management resources focus intensely on the patients most likely to return, the 11.47% who generate our penalties rather than spreading those resources inefficiently across our entire patient population. Given that the High-Risk group has a 17.1% readmission rate, interventions targeted at this group have the highest return on investment.

6.2 Recommendation 2: The "Home Health" Safety Net Audit

Given the alarming 54% readmission probability for patients discharged to Home with Home Health Services, we need to launch an immediate quality improvement audit of our home health partner agencies and our internal protocols for this transition.

We propose a new policy requirement: Any patient flagged as High Risk by VCI who is being discharged to home with home health must have a confirmed nursing visit scheduled within 24 hours of arrival home, not the current standard of 48-72 hours. Additionally, we should require a standardized handoff call between the VHN discharge nurse and the home health agency nurse before the patient leaves the building, ensuring clear communication about high-risk medications, warning signs to watch for, and the specific interventions needed.

The data proves this discharge pathway is our largest operational leak, the place where over half of these patients are falling through the cracks despite our best intentions. By tightening the safety net specifically for this transition, we can plug the gap where we're currently losing patients.

6.3 Recommendation 3: Targeted "Insulin Mastery" Education

Discontinue our generic discharge education approach for insulin-dependent patients. Instead, implement a mandatory "Insulin Mastery" educational track that these patients must complete before discharge.

This specialized track should include hands-on demonstration and return demonstration of injection technique, comprehensive review of hypoglycemia recognition and treatment, discussion of insulin storage requirements, and assessment of the patient's home environment for potential barriers to compliance. We should implement a "teach-back" requirement where patients must successfully demonstrate insulin administration back to the nurse before being medically cleared for discharge.

Our data clearly identifies insulin users as our highest clinical risk cohort, with readmission rates approaching 49%. Treating them with the same generic diabetes education we give to everyone else is failing both them and our institution. Targeted, intensive education that specifically addresses the

complexities of insulin management tackles the root cause of their elevated readmission risk: medication mismanagement and complications arising at home.

7. Conclusion

VHN possesses all the data necessary to solve our HRRP challenge, we simply need the right analytical lens to extract actionable insights. The patterns were always present in our electronic records, waiting to be discovered. By adopting the Vitality Complexity Index as a standard clinical tool and addressing the specific operational blind spots we've identified in home health transitions and insulin patient education, VHN can achieve measurable reductions in readmission penalties, improve our financial margins, and most importantly, fulfill our fundamental mission: ensuring our patients stay healthy at home rather than cycling repeatedly through our emergency department.

The path forward is clear, evidence-based, and immediately implementable. The question isn't whether we can reduce readmissions, the data proves we can. The question is whether we have the organizational will to act on what the evidence is telling us.